

Booking SaaS

Phased Build Plan — Phase 0 through Phase 3

Studio management for ceramics studios — classes, wheels, kilns and pieces in one system.
A multi-tenant scheduling platform covering fixed-venue and mobile delivery, built for the United States market on a self-serve model.

Market	United States
Beachhead vertical	Pottery & ceramics studios
Go-to-market	Pure self-serve SaaS
Foundation	saas-starter — Express + TypeScript + Postgres/Prisma
Origin reference	FDG WP Booking Flow plugin (design reference only)
Total duration	19 weeks to a sellable product, plus ongoing Phase 3
Document	v1 — 4 August 2026

Programme overview

The programme is sequenced so that correctness is established before surface area. Phase 0 produces no user-visible output at all — it exists to make overbooking, timezone drift and resource collisions structurally impossible before a single screen is built. Every later phase depends on that foundation and none of it is revisited.

Each phase closes on an **exit gate**: a small set of objectively verifiable conditions. A phase is not complete because its tasks are ticked off; it is complete when its gate passes. This is the single most important discipline in the plan.

Phase	Focus	Duration	Outcome
Phase 0	Scheduling core (headless)	3 weeks	A provably correct engine. No UI.
Phase 1	Self-serve MVP	9 weeks	A stranger signs up and takes a real booking unaided.
Phase 2	Vertical depth	7 weeks	Courses, pieces and kilns. Becomes a category product.
Phase 3	Scale and expansion	Ongoing	API, reporting, second vertical.

Timeline

When	Stage	Activity
Weeks 1–3	Phase 0	Scheduling core, concurrency, time handling
Weeks 4–12	Phase 1	MVP build; SEO content starts week 4; 10DLC filed week 4
Week 12	Milestone	Public launch. Trials open, marketing site live
Weeks 13–19	Phase 2	Courses, attendance, pieces, kilns, widget, WP plugin
Week 19	Milestone	Category-complete product. Full-price tiers justified
Week 20+	Phase 3	API, reporting, additional calendars, second vertical

Long lead-time items — start these in week 1

Three dependencies have external approval queues that are longer than the phases that need them. Filing late is the most common avoidable delay in this plan.

Item	Lead time	Needed by	Action
Stripe Connect platform application	1–3 weeks	Phase 1, week 6	Apply in Phase 0, week 1
Twilio A2P 10DLC brand + campaign	2–6 weeks	Phase 1, week 9	File in Phase 0, week 1

Item	Lead time	Needed by	Action
Domain, name and trademark check	—	Phase 1, week 4 (SEO)	Decide before Phase 0 ends

Cross-phase workstreams run continuously from week 4 onward and are not listed separately in each phase: SEO content production, observability and alerting, customer support tooling, and security/compliance hygiene (GDPR-grade data export and deletion, audit logging, TCPA consent capture).

PHASE 0

Scheduling Core

3 weeks · Headless · No user interface

Objective

Build and prove the scheduling engine in isolation. Every hard correctness problem in booking software — concurrent seat allocation, overlapping appointments, recurrence across daylight-saving boundaries, resource contention — is solved here, at the database layer, before any interface exists to obscure it.

This phase is the one most likely to be rushed and the one where rushing is most expensive. The original WordPress plugin shipped a check-then-insert capacity check and a unique index that only caught exact start-time collisions. Both work fine for one studio and fail predictably under real concurrency. That class of defect cannot be patched later — it has to be designed out.

Workstreams

ID	Workstream	Key tasks	Est.
0.1	Project foundation	- Fork saas-starter; strip to the clean-architecture skeleton - Postgres 16 with PostGIS and btree_gist extensions - Docker Compose: api, postgres, redis - CI pipeline running the full test suite on every commit - Structured logging and error handling baseline	2 days
0.2	Schema and migrations	- Prisma schema for all core entities; organization_id on every table - Composite unique constraints scoped to organization_id throughout - Tenant isolation enforced at the repository layer, not per-query - Seed script producing a realistic ceramics studio fixture - Migration strategy and rollback procedure documented	3 days
0.3	Time and recurrence	- All instants stored as timestampz; IANA timezone on every location - Wall-clock intent preserved alongside the UTC instant - iCal RRULE expansion evaluated in local zone, then converted - Availability overrides: day off, custom hours, extra hours - Explicit DST resolution policy for non-existent and ambiguous local times	4 days
0.4	Availability engine	- Candidate staff resolution by qualification and location coverage - Rule expansion, override application, break subtraction - Busy-interval assembly from bookings and synced calendar blocks - Slot emission at per-service granularity with padding and notice windows - Resource availability check (wheels, kilns) within the candidate window - Travel-time widening for mobile bookings; haversine pre-filter plus cached matrix	5 days

ID	Workstream	Key tasks	Est.
0.5	Concurrency and integrity	• EXCLUDE USING gist constraint preventing staff appointment overlap • Row-locked seat counter with CHECK constraint for sessions • EXCLUDE constraint on resource allocations • Booking holds with expiry, plus the background release job • Idempotency keys on all mutating operations	4 days
0.6	Verification harness	• Concurrency load test: N parallel requests against the final seats • DST test suite across at least two US timezones, both transitions • Availability performance benchmark over a 31-day window • Cross-tenant isolation test asserting zero data leakage	2 days

Deliverables

- Versioned Prisma schema and migration set for the full core domain
- Availability engine exposed as an internal service module with a stable contract
- Three enforced concurrency mechanisms, all at the database layer
- Automated test suite covering concurrency, DST, isolation and performance
- Written architecture decision records for time handling and concurrency

Exit gate — all four must pass before Phase 1 begins

- **Zero overbooking.** 100 concurrent requests for the final 5 seats produce exactly 5 bookings and 95 clean, correctly-typed failures.
- **DST correctness.** A weekly recurring class holds its local wall-clock time across both spring-forward and fall-back, in at least two US timezones.
- **Performance.** Availability for a 31-day window returns at p95 under 200ms with no external network call on the read path.
- **Tenant isolation.** The isolation suite reports zero cross-organization reads or writes under adversarial input.

PHASE 1

Self-Serve MVP

9 weeks · Public launch at close

Objective

Turn the engine into a product a stranger can buy without ever speaking to a human. Because the go-to-market is pure self-serve, onboarding quality and search visibility are not polish items deferred to a later phase — they are the delivery mechanism for the entire business and ship inside this phase.

Workstreams

ID	Workstream	Key tasks	Est.
1.1	Auth and tenancy	• Registration, login, refresh tokens, password reset, email verification • Organization creation, membership roles: owner, admin, instructor, front desk • Role-based permission checks across every endpoint	1 wk
1.2	Admin CRUD	• Service types with pricing, duration, capacity, deposit and policy settings • Staff management, service qualifications, working hours and time off • Locations: fixed venues and mobile service areas with radius and travel fee bands • Cancellation and refund policies stored as configurable data, not code	1.5 wk
1.3	Public booking page	• Server-rendered per-organization page for search visibility and speed • Flow: location, service, staff or seats, date and time, details, payment • Address capture and venue-requirements step for mobile bookings • Per-organization branding; fully responsive; accessible • Customer-facing cancel and reschedule via secure random token	2 wk
1.4	Payments	• Stripe Connect Express onboarding inside the setup wizard • Deposit and full-payment flows; server-computed amounts only • Booking hold reserved through checkout, released automatically on abandon • Refunds honouring the configured cancellation policy • Webhook handling for asynchronous payment outcomes	1.5 wk
1.5	Notifications	• Transactional email: confirmation, reminder, cancellation, reschedule • SMS reminders via Twilio with TCPA consent capture and opt-out handling • Queued delivery with retries and per-organization templates • A2P 10DLC campaign registration completed and verified	1 wk
1.6	Calendar sync	• Per-staff Google OAuth connection • Outbound event create, update and delete on booking changes • Inbound push notifications writing to the busy-blocks mirror • Loop guard so self-authored events never block their own sessions	1 wk

ID	Workstream	Key tasks	Est.
1.7	Owner dashboard	• Today view: schedule, arrivals, alerts • Calendar: day, week and month with drag-to-reschedule • Bookings list with search, filters, bulk actions and manual booking entry • Customer records with booking history	1.5 wk
1.8	Onboarding and billing	• Setup wizard: studio and timezone, Stripe, first class, hours, publish • Pottery-specific seeded defaults so studios edit rather than create • Stripe Billing subscription, 14-day no-card trial, plan limit enforcement • Dunning, upgrade and downgrade flows	1 wk
1.9	Marketing site and SEO	• Positioning site with pricing, feature and comparison pages • Competitor alternative pages: Momenze, Sawyer, Punchpass • Initial long-form content targeting studio-owner search intent • Analytics, conversion tracking and technical SEO foundations	1 wk

Workstreams 1.9 and parts of 1.5 run in parallel with development, which is why the sum of estimates exceeds the nine-week calendar duration.

Deliverables

- A live, publicly available product with self-serve signup and paid subscriptions
- Hosted booking page per studio, covering both fixed-venue and mobile bookings
- Stripe Connect payments with deposits, refunds and policy enforcement
- Email and SMS notification pipeline, 10DLC approved
- Two-way Google Calendar synchronisation
- Marketing site with the first tranche of SEO content published

Exit gate

- **Unaided activation.** Three external testers, with no help from the team, sign up and publish a live booking page in under ten minutes each.
- **Real money.** A genuine booking is taken and settled through Stripe Connect into a studio's own account.
- **Notification reliability.** Confirmation and reminder delivery exceeds 99% across email and SMS over a one-week soak test.
- **No manual intervention.** A complete signup-to-first-booking journey occurs with zero human contact from the team.

PHASE 2

Vertical Depth

7 weeks · The differentiation phase

Objective

Phase 1 produces a competent, cheaper booking tool. Phase 2 produces something no competitor offers. Ceramics studios have three operational realities that generic scheduling software does not model at all: multi-week courses with make-up sessions, the two-to-four week firing cycle that governs when a customer gets their work back, and kilns and wheels as physically constrained resources.

Each of these is a feature on its own. Delivered together, alongside the mobile travel engine from Phase 1, they constitute a category rather than a cheaper alternative — and they are what justifies the upper pricing tiers.

Workstreams

ID	Workstream	Key tasks	Est.
2.1	Course series and cohorts	• Series as a first-class entity: book once, pay once, attend many • Cohort scheduling with automatic session generation from a series template • Single-transaction enrolment across every session in the series • Roster views, capacity managed at series and session level • Prerequisites and skill-level gating between course tiers	2 wk
2.2	Attendance and make-ups	• Per-session attendance marking: attended, absent, no-show • Automatic make-up credit issuance on a recorded absence • Credit redemption against an equivalent session in another cohort • Credit expiry rules and customer-visible credit balance	1 wk
2.3	Piece tracking	• Pieces created against a booking or entered manually, with photo and shelf code • Lifecycle: drying, bisque queued, bisque fired, glazing, glaze queued, glaze fired, ready, collected • Batch state transitions for a whole kiln load • Automated pickup SMS the moment a piece reaches ready • Customer-facing lookup so students can check status themselves	1.5 wk
2.4	Kiln and firing management	• Kilns as blocking resources across a full firing and cooldown cycle • Firing queue and load builder; pieces assigned to a firing • Conflict prevention against classes and other firings • Firing history and per-kiln utilisation view	1.5 wk
2.5	Resources	• Wheels, slab rollers, kits and vehicles as counted resources • Service-to-resource requirements driving true capacity ceilings • Concurrent-allocation checks folded into the availability engine	0.5 wk

ID	Workstream	Key tasks	Est.
2.6	Waitlists and packages	• Waitlist entries with position and automatic promotion on cancellation • Time-boxed claim window for a released seat • Class packs and credit bundles with expiry • Open-studio membership hours distinct from scheduled classes	1 wk
2.7	Distribution surfaces	• Embeddable booking widget: single script tag, iframe isolated • WordPress plugin as a thin shim, published free on the plugin directory • Squarespace and Wix embed instructions and snippets	1 wk

Deliverables

- Multi-week course enrolment with attendance tracking and make-up credits
- End-to-end piece lifecycle with automated pickup notifications
- Kiln scheduling integrated with class scheduling and resource constraints
- Waitlists, class packs and open-studio memberships
- Embeddable widget plus a published WordPress distribution plugin

Exit gate

- **Full course cycle.** A studio runs a complete six-week course: enrolment, attendance across all sessions, at least one absence, and a redeemed make-up credit.
- **Full firing cycle.** A piece is tracked from creation through both firings to collection, with the pickup notification delivered automatically.
- **No scheduling conflicts.** Kiln and class scheduling coexist for a full month with zero resource collisions.
- **Distribution live.** The WordPress plugin is approved and listed, and the widget is embedded on at least one live customer site.

Phase 2 scope is fixed: courses, pieces, kilns, resources, waitlists, packages and distribution. Nothing else. This is the phase where scope creep is most tempting and most damaging, because every studio interviewed will request something adjacent.

PHASE 3

Scale and Expansion

Ongoing · Prioritised by customer demand

Objective

With a category-complete product and paying customers, Phase 3 shifts from building what the product needs to building what customers pull. The sequence below is a default ordering, not a commitment; real usage data and churn reasons should reorder it. The final item — the second vertical — is the one that decides whether this becomes a large-market business or a good niche one.

Workstreams

ID	Workstream	Key tasks	Est.
3.1	Public API and webhooks	• Documented REST API with per-organization keys and rate limits • Outbound webhooks for booking, payment and piece lifecycle events • Zapier and Make integrations built on the public API • Unlocks larger multi-site customers and partner integrations	3 wk
3.2	Reporting and analytics	• Revenue by service, instructor, location and channel • Class occupancy and utilisation; no-show and cancellation rates • Course completion and student retention cohorts • Mobile-booking profitability including travel cost • Scheduled exports and email digests	2 wk
3.3	Additional calendars	• Microsoft Graph two-way sync for Outlook • Apple Calendar via CalDAV • Unified connection management in staff settings	2 wk
3.4	Commerce expansion	• Gift cards and gift certificates • Recurring memberships with billing and access rules • Retail and materials sales attached to a booking • Tipping and gratuity	3 wk
3.5	Marketplace and discovery	• Public directory of studios and classes with search by location and date • SEO-optimised class landing pages generating demand for tenants • Turns the platform from a cost centre into a lead source • Significant strategic decision — validate demand before building	4 wk
3.6	Second vertical	• Glass blowing, woodworking, jewellery and candle making share the same shape: constrained equipment, curing or firing cycles, multi-week courses • Engine is already horizontal; work is templates, defaults, copy and SEO • Target roughly 4 weeks per additional vertical once the first is proven	4 wk

Sequencing guidance

Do not start Phase 3 on schedule. Start it on evidence. The correct trigger is a clear signal from paying customers — churn attributed to a missing capability, repeated sales objections, or upgrade requests blocked by a specific gap. Building 3.5 before that signal exists is the most expensive mistake available in this phase.

Risk register

Risk	Severity	Mitigation
Self-serve funnel produces no revenue for months	High	SEO and content begin in Phase 1 week 4; measure activation, not signups
Scope creep during Phase 2	High	Fixed scope list; adjacent requests logged to Phase 3, never absorbed
Concurrency defects reach production	High	Database-level constraints plus the Phase 0 gate test; never trust app logic
Timezone and DST bugs in a multi-timezone market	Medium	Explicit DST suite in Phase 0; timestamptz and wall-clock intent throughout
Building blind without customer contact	Medium	Interview three US studios before Phase 0 closes, even with no sale attached
A2P 10DLC approval delays SMS launch	Medium	File in Phase 0 week 1; email-only reminders as the fallback path
Incumbents add ceramics-specific features	Low	Their models are fitness-shaped and the segment is too small to attract them

Immediate next steps: decide the product name and secure the domain; file the Stripe Connect and Twilio 10DLC applications; book interviews with three US ceramics studios; and begin Phase 0 with the Prisma schema and the concurrency test harness.